
SULPHONIX H₂S DOSIMETER WRISTBAND

Passive Colorimetric Exposure-Dosimeter with Quantitative Reading

Field Operating Manual, Worker Safety Guidelines, and Chemical Exposure Protocols for Petroleum Refineries, Hydrotreaters, and Hazardous Processing Units.

REGISTERED TEAM: SULPHONIX

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Wearable Field Exposure Dosimeter • Operations & Safety Reference

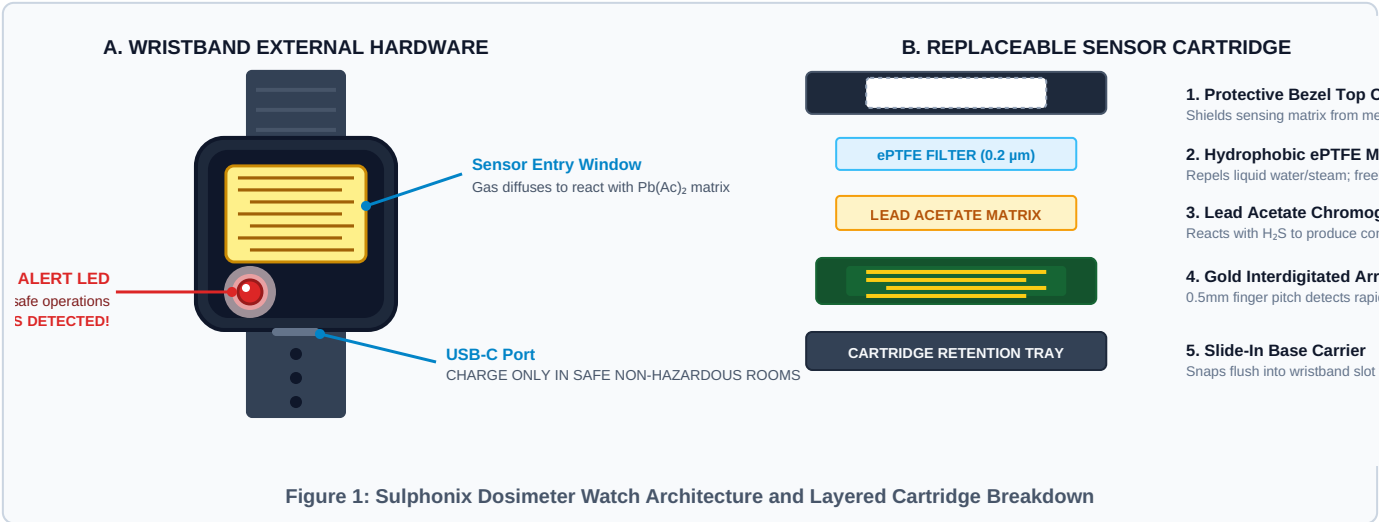
CRITICAL SAFETY ALERT: NEVER RELY ON YOUR SENSE OF SMELL!



Above **100 PPM**, Hydrogen Sulfide (H₂S) induces instant **olfactory paralysis (total loss of smell)**. You will smell **NOTHING** even in fatal concentrations. The wristband LED is engineered to **ONLY GLOW RED WHEN DANGEROUS GAS LEVELS ARE DETECTED**. When it glows, evacuate immediately!

1. WRISTBAND LAYOUT & REPLACEABLE CARTRIDGE ANATOMY

Review the physical interface and internal cartridge layers before commencing shift operations:



2. DANGER-ONLY LED ALERT LOGIC & WORKER ACTION

The Sulphonix wristband operates on a **Silent/Dark Baseline**: during safe plant operations, the LED remains completely **OFF** to prevent visual fatigue. It **ONLY GLOWS WHEN HAZARDOUS LEVELS OCCUR**.

LED Appearance	H ₂ S Exposure Level	What This Means (Hazard State)	MANDATORY WORKER ACTION
LED IS COMPLETELY OFF	< 10 PPM (Normal Safe Air)	Safe work atmosphere. Normal background baseline. Dosimeter is quietly measuring.	Continue assigned tasks. Wear standard plant PPE (hard hat, safety glasses, coveralls, boots).
PULSING RED GLOW	10 – 50 PPM (Danger Threshold)	OSHA Permissible Ceiling Exceeded! Eye burning, coughing, headache will begin.	STOP WORK IMMEDIATELY! Walk briskly UPWIND. Report leak immediately to the control room / unit operator.
INTENSE SOLID RED + CONTINUOUS BUZZ	> 50 – 100+ PPM (LETHAL / IDLH)	IMMINENT LIFE THREAT! Olfactory fatigue occurs instantly (NO SMELL). Risk of immediate knockdown and asphyxiation.	RUN UPWIND IMMEDIATELY! Don Self-Contained Breathing Apparatus (SCBA). NEVER attempt to rescue a fallen buddy without SCBA!

OPERATING RULES: DOS AND DON'TS

Proper Field Handling Protocols for Maximum Safety & Accuracy

3. WHAT TO DO WITH THE WATCH (STANDARD OPERATING WORKFLOW)

RULE 1 • PRE-SHIFT INSPECTION

Confirm the watch battery is adequately charged. Check that the cartridge slot is free of debris, dust, or oil before sliding in a fresh cartridge.

RULE 2 • INSERT CARTRIDGE UNTIL IT CLICKS

Open a hermetically sealed Sulphonix cartridge. Check that the paper is white/cream. Slide it straight into the slot until an audible mechanical **CLICK** confirms lock.

RULE 3 • SYNC WORKER ID VIA NFC / BLE

Open the companion mobile app and tap the phone to the watch face to link your worker ID badge. This locks baseline zero-exposure calibration for your 8-hour shift.

RULE 4 • WEAR WATCH OUTSIDE SLEEVES & PPE

Fasten the strap snugly over your coverall cuff. The sensor window must always face outwards toward ambient air and never be obscured by gloves or coats.

RULE 5 • TRUST THE GLOWING LED — EVACUATE!

The moment the LED pulses or shines red, drop non-critical tools and retreat upwind immediately. Never hesitate or wait for a smell to confirm danger!

RULE 6 • SCAN BADGE DURING REST BREAKS

During designated meal or rest breaks, capture a quick photo with the Sulphonix app to calculate and log your accumulated cumulative dosage (PPM-hr).

4. WHAT NOT TO DO WITH THE WATCH (CRITICAL PROHIBITIONS)

✗ PHYSICAL & HARDWARE DON'TS

- **NEVER Cover the Sensor Window:** Never apply tape, stickers, protective films, or cover with bulky rubber sleeves. If gas cannot reach the strip, you have ZERO protection!
- **NEVER Puncture the ePTFE Filter:** Do not probe the sensor window with pens, wire, or screwdrivers. Puncturing the membrane lets moisture short the electrodes.
- **NEVER Charge in Hazardous Plant Units:** Charging via USB-C can generate tiny micro-sparks. Recharge ONLY in designated non-hazardous electrical bays/control rooms.
- **NEVER Submerge in Liquids:** Avoid dropping into drain trenches, oily condensate, or chemical wash buckets. The hydrophobic filter resists light splash, not immersion.

✗ PROCEDURAL & BEHAVIORAL DON'TS

- **NEVER Re-Use a Discolored Cartridge:** If a cartridge has turned brown or metallic black, its lead acetate is fully spent. Never carry it into another shift!
- **NEVER Ignore a Glowing Red LED:** Do not silence or bypass the alert. Olfactory paralysis may make the area feel completely safe when it is fatal.
- **NEVER Discard Cartridges in Regular Bins:** Used cartridges contain trace lead compounds. Always deposit spent units in yellow Heavy-Metal Hazardous Disposal receptacles.
- **NEVER Attempt Solo Rescues:** If a co-worker collapses in a red-alert area, DO NOT charge in to pull them out without SCBA. Over 60% of H₂S fatalities are rescuers!

5. SHIFT TIMELINE FOR FIELD WORKERS



Figure 2: Routine Shift Lifecycle for Plant Workers using Sulphonix Wristbands

DEVICE ENVIRONMENTAL & SAFETY GUIDELINES

Sunlight Protection, Storage Limits, Age Restrictions & Handling Precautions

6. SUNLIGHT, HEAT & ENVIRONMENTAL PROTECTION GUIDELINES

SUNLIGHT & UV DEGRADATION WARNING

Protect from Prolonged Direct Sunlight: Unexposed lead acetate reagent matrices slowly photodecompose under intense UV solar radiation, causing superficial yellow-browning and premature electrical drift.

Field Practice: When working outdoors during midday, keep the watch facing sideways or shaded under normal body posture. Never leave the watch on vehicle dashboards, sunny window sills, or outdoor lockers when off-shift.

STORAGE TEMPERATURE & HUMIDITY SPECS

Storage Temperature: Store unused cartridges between 10°C and 30°C in their original airtight desiccated foil pouches. Avoid hot toolboxes (> 45°C).

Operating Range: The wristband operates from -10°C to 55°C. In freezing ambient weather, avoid direct ice accumulation on the ePTFE gas aperture.

CARTRIDGE SHELF LIFE & HANDLING

Sealed Shelf Life: 12 months from manufacturing date when kept cool and dry in original foil.

Post-Unsealing Rule: Once the foil packet is torn, insert the cartridge into the watch immediately. Unsealed cartridges exposed to humid ambient air should be consumed within 8 to 12 hours.

Chemical Safety: The cartridge contains bonded lead acetate (\$Pb(CH_3COO)_2\$). Do not touch the inner reagent layer with bare hands, chew, or allow to contact open wounds.

CLEANING & DECONTAMINATION PROTOCOLS

Outer Casing: Clean silicone straps and external plastic with mild damp wipes or 70% isopropyl alcohol wipes.

DO NOT Wipe the Sensor Window: Never wipe, rub, or spray liquid cleaning solutions directly across the white ePTFE membrane. Doing so forces solvents into the reactive pad and permanently ruins calibration.

7. USER ELIGIBILITY, AGE RESTRICTIONS & REGULATORY APPROVALS

18 AGE RESTRICTION & EMPLOYMENT ELIGIBILITY

- Strict 18+ Adult Use Only:** The Sulphonix dosimeter wristband is certified strictly as an **Industrial Safety Wearable** for adult chemical and refinery workers aged 18 years and above.
- Not a Consumer Smartwatch:** This device is NOT a lifestyle, fitness, medical, or pediatric monitoring gadget. Keep out of reach of children and unauthorized domestic individuals.
- Pregnancy & High-Susceptibility Advisories:** Expectant female personnel must consult plant occupational health physicians prior to entering sour gas units, as fetal susceptibility to hypoxia from H₂S exposure is acute.
- Trained Personnel Mandatory:** Only workers who have completed mandatory plant hazardous gas training (OSHA / OISD-114 certified) are permitted to wear and operate this dosimeter in active refinery battery units.

⚡ INTRINSIC SAFETY & ELECTRONIC CARE

- Zone 1 / Zone 2 Certified:** The internal electronics (ESP32-C3, battery current limiter) are engineered to meet intrinsically safe hazardous area ratings when the casing is fully intact.
- Damaged Casing Policy:** If the wristband is dropped, crushed, or shows cracked glass or housing, remove it from service immediately. Damaged casings void intrinsic safety approvals!
- No User-Serviceable Internal Parts:** Do NOT attempt to unscrew the rear chassis or replace the lithium-polymer cell. Tampering voids warranty and creates an ignition hazard in volatile plant units.
- Recharging Isolation:** Never connect charging adapters while in the field. Charge only in designated, positive-pressure safe control rooms with certified chargers.

8. QUICK REFERENCE: DEVICE OPERATIONAL LIMITS & RATINGS

Parameter	Permissible Operational Limit	Field Precautions & Worker Guidance
Operating Ambient Temp	-10°C to +55°C (14°F to 131°F)	Avoid touching scorching steam pipes (> 70°C) with the wristband casing.
Storage Temperature	+10°C to +30°C (cool, dry storage)	Do not store in direct sun or high-temperature vehicle interiors.
Relative Humidity Range	5% to 95% RH (Non-condensing)	Built-in 0.2 µm ePTFE filter repels water spray and monsoon showers.
Ingress Protection	IP67 Equivalent (Splash & Dust Resistant)	Resistant to high dust, rain, and mud. Submersion in hydrocarbons prohibited.
Minimum Worker Age	18 Years (Certified Adult Operators)	Factory Act 1948 & OSHA workplace hazardous chemical worker criteria.

H₂S HAZARD LADDER & EVACUATION SURVIVAL

Medical Hazard Data, Refinery Danger Zones & Evacuation Action Plan

9. H₂S CONCENTRATION HAZARD LADDER & HUMAN IMPACT

Concentration	Human Physiological Symptoms	Watch LED State	Required Emergency Action
0.01 – 0.3 PPM	Noticeable odor of rotten eggs. Completely non-toxic.	LED IS OFF	Normal plant operation. dosimeter remains idle/passive.
1.0 – 5.0 PPM	ACGIH 8-Hour TWA limit (1 ppm) & STEL limit (5 ppm).	LED IS OFF	Safe for 8-hr shifts. dosimeter tracks cumulative dosage.
10 – 20 PPM	OSHA Permissible Ceiling (20 ppm). Eye burning, coughing, headache.	PULSING RED	LEAVE THE SPOT! Inform supervisor. Locate fugitive flange leak.
100 PPM (IDLH)	IMMEDIATE OLFACTORY PARALYSIS (Smell disappears). Stinging eyes, severe lung damage.	INTENSE RED GLOW	EVACUATE IMMEDIATELY! Do not assume gas cleared because smell stopped!
300 – 500 PPM	Severe pulmonary edema. Pulmonary hemorrhaging, collapse within 15–30 min.	INTENSE RED + BUZZ	FULL EVACUATION. Rescue personnel must wear Positive-Pressure SCBA only.
700 – 1000+ PPM	INSTANT KNOCKDOWN. Rapid unconsciousness in 1–2 breaths. Death in minutes.	INTENSE RED + BUZZ	FATAL ZONE. Automated plant unit shutdown and water curtains triggered.

10. EMERGENCY EVACUATION: THE WINDSOCK & CROSSWIND RULE

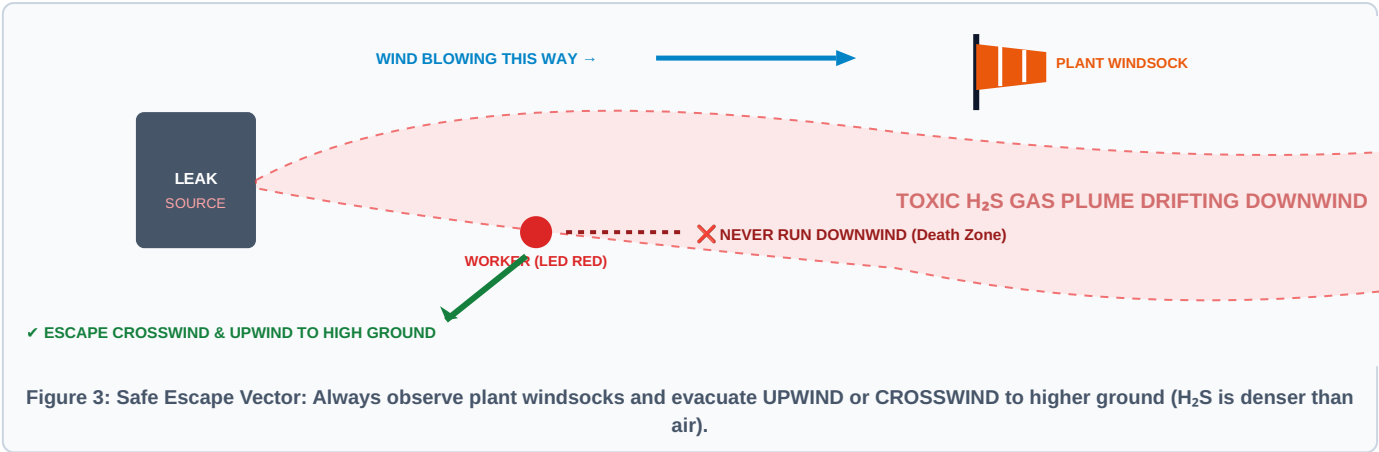


Figure 3: Safe Escape Vector: Always observe plant windsocks and evacuate UPWIND or CROSSWIND to higher ground (H₂S is denser than air).

11. HIGH-RISK REFINERY HOTSPOTS & THE BUDDY SYSTEM RULE

REFINERY VULNERABILITY HOTSPOTS

- Sulphur Recovery Unit (SRU):** Stripper gaskets & tail gas release > 500 ppm pure H₂S.
- Hydrotreater & DHDS Units:** High-pressure separator flanges and compressor seals.
- Delayed Coker Unit (DCU):** High fugitive emission during coke drum unheading.
- Effluent Treatment Plant (ETP):** Sour trenches hide heavy, stagnant gas pockets.

THE STRICT "BUDDY SYSTEM" RULE

- NEVER enter hazardous battery limits alone.** Always maintain continuous visual and vocal contact with your assigned buddy.
- IF YOUR BUDDY COLLAPSES:**
- DO NOT** run into the gas cloud to drag them out without SCBA.
 - Sound the emergency plant alarm immediately.
 - Only trained rescue squads equipped with positive-pressure SCBA may enter.